

A Predictability Certificate for Equivariant World Models

Scale buys interpolation, structure buys a certificate — across configuration, horizon, and resolution

Hongbo Wang

Department of Mathematics, Stony Brook University, Stony Brook, NY 11794, USA

△ **FOR REVIEW — structural choices flagged (delete this block before any submission)**

This is a **new, additive** first draft of *paper2* — the certificate paper — assembled from the proposal *paper2*-certified-compositional-jepa and the eight experiments that now back it (Steps 47, 49, 50, 51, 52, 53, 57, 58). **Nothing else was touched**: the three v1/v2 papers (equivariance_generalization_core, geometric_payoff, equivariant_lejepa, main_compact) and the frozen arXiv bundles are undisturbed. Wiring this into `arxiv/build.py` is a separate, explicit step you can greenlight later.

Judgment calls for you to accept or override: 1. **Title** — chose “A Predictability Certificate for Equivariant World Models” from the proposal §9 menu. Alternatives there: “*Scale Buys Interpolation, Structure Buys a Certificate*”, “*The Certified Region of an Equivariant World Model*”. Easy to swap. 2. **Scope of empirical claims** — every number below is from a committed experiment with a passing seed-gate (3 seeds where noted) and, for the load-bearing ones, a unit test. The toy scale (1–2 GPU / CPU) is stated openly; this is a *mechanism* paper, not a scaled-SOTA benchmark (proposal §7). 3. **The Noether hinge (§3/§5.3)** is presented as a **measured conjecture, now confirmed on a controlled $SO(2)$ system**, with the honest non-converse (invariant $\nRightarrow$ slow). The open part — lifting it to the *approximate*-symmetry embodied model — is flagged as future work, not hidden. 4. **Relationship to the old paper** — the old paper’s $[A]/[B]/[C]$ device (equivariance $\Rightarrow$ constant error / one-step / closed-loop) is the **single-element, single-resolution corner** of this paper’s certificate; stated explicitly in §6 so the two papers don’t appear to overclaim the same thing.

Abstract

Scaling a world model lowers test error on the data manifold but issues **no guarantee about any specific unseen situation**. We give a different *kind* of result for **equivariant** latent world models — a **predictability certificate**: a provable, computable region of situations the model is guaranteed to handle *without further training or data*. The certificate spans three orthogonal axes at once: **configuration** (the exponentially large monoid $\langle S \rangle$ generated by k primitive symmetries, certified by checking the k generators), **horizon** T (how far ahead the guarantee reaches), and **resolution** ϵ (at what level of detail). A master theorem makes this precise: under exact equivariance with an orthogonal latent representation the whole-pipeline error is *invariant* over all of $\langle S \rangle$ (Theorem A), and a spectral refinement (Theorem B) shows each latent channel is certified to a horizon $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ set by its Lyapunov exponent — so the certified region is the **coarse-invariant, slow, low-composition** corner. We confirm all three axes empirically at small scale: (i) a model trained on 6 generators of $\mathbb{Z}_2^6$ is certified over all $2^6=64$ compositions to machine zero while a matched MLP degrades monotonically; (ii) a learned predictor recovers the chaotic Lyapunov exponent to within $\sim 0.4\%$ (3 seeds) and its certified horizon obeys the $\log(1/\epsilon)$ law; (iii) on an $SO(2)$ mechanical system the conserved (slowest) quantities organize into the architecturally invariant channels — a measured **Noether hinge** linking the symmetry axis to the time axis; and (iv) a non-equivariant MLP **scaled across an $88\times$ parameter range** buys in-distribution interpolation

— even beating the equivariant model there — yet over the unseen orbit the best-scaled MLP stays 10–155× above the equivariant floor (across 3 seeds) and never reaches it. *Scale buys interpolation; structure buys a certificate.* The contribution is a tool: a single criterion for *what* an equivariant world model can certifiably predict, and a structural reading of *why* celestial mechanics is predictable for millennia while weather dies at two weeks.

1. Introduction

Two debates organize modern world-model research. The first is *generation vs. abstraction*: pixel-space simulators (diffusion / video models) against latent predictive models (JEPA) that predict representations rather than pixels. The second is *scale vs. structure*: the Bitter-Lesson view that brute-force scaling eventually wins, against the geometric-deep-learning view that symmetry priors buy sample efficiency. Both debates are usually run as **degree** contests — who needs fewer pixels, who needs less data.

We argue the more useful question is a **kind** contest. Scaling produces a model with low *average* error on the data distribution, but it cannot certify anything about a *named, specific* situation the data did not cover. Structure can. For an equivariant world model we can write down, in advance and without further data, the exact set of situations on which the model’s behaviour is **guaranteed** — a *predictability certificate*. This reframes “is structure worth it?” from a benchmark race to a question about *what kind of statement each approach can make*.

The certificate has three orthogonal axes:

- **Configuration** $w \in \langle S \rangle$: the model generalizes across the (exponentially large) monoid generated by k primitive symmetries $S = \{g_1, \dots, g_k\}$, and we *certify the whole monoid by checking only the k generators*.
- **Horizon** T : how far into the future the guarantee reaches.
- **Resolution** ϵ : at what level of detail the guarantee holds.

Our contributions:

1. **A three-axis master theorem** (§2): composition closure (k checks $\Rightarrow$ exponential set), an exact certificate under exact equivariance (Theorem A), and a spectral degradation law $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ (Theorem B) whose certified region is the coarse-invariant-slow-low- $|w|$ corner.
2. **The Noether hinge** (§3): the conjecture — *measured and confirmed on a controlled system* — that the group-invariant channels are the dynamically slow (long-horizon-certifiable) ones, linking the configuration axis to the horizon axis.
3. **Empirical confirmation of all three axes at small scale** (§4), including the headline contrast: *scale buys interpolation; structure buys a certificate*.

We are explicit about scope (§6): this is a mechanism/theory paper with 1–2-GPU proof-of-principle, not a scaled-SOTA benchmark, and the hinge’s lift to *approximate* symmetry is open.

2. Setup and the three-axis master theorem

Setup and assumptions. An encoder $E : \mathcal{X} \rightarrow \mathcal{Z}$ and an action-conditioned predictor $f : \mathcal{Z} \times \mathcal{A} \rightarrow \mathcal{Z}$; the true environment transition is $\Phi : \mathcal{X} \times \mathcal{A} \rightarrow \mathcal{X}$. A finite generating set $S = \{g_1, \dots, g_k\}$ generates a monoid $\langle S \rangle$ whose elements (words $w = g_{i_1} \dots g_{i_m}$, length $|w| = m$) act on states by $g \cdot x$, on latents by a representation $\rho(g)$, and on actions by $\sigma(g)$. Write $f^{(T)}(z; \vec{a})$ for the T -step latent rollout under actions $\vec{a} = (a_1, \dots, a_T)$, x_T^{true} for the T -step true state, and $\text{Err}_T(x; \vec{a}) = \|f^{(T)}(E(x); \vec{a}) - E(x_T^{\text{true}})\|$ for the whole-pipeline prediction error, where $\|\cdot\|$ is the Euclidean norm on $\mathcal{Z}$ (the experiments use relMSE, its square). We use four assumptions, each *checkable on the k generators*:

- **(A1) Encoder equivariance:** $E(g_i \cdot x) = \rho(g_i) E(x)$.
- **(A2) Predictor equivariance:** $f(\rho(g_i)z, \sigma(g_i)a) = \rho(g_i)f(z, a)$.
- **(A3) The group is a symmetry of the dynamics:** $\Phi(g_i \cdot x, \sigma(g_i)a) = g_i \cdot \Phi(x, a)$ — so rolling out the *transformed* state equals transforming the *rolled-out* state, $(g_i \cdot x)_T^{\text{true}} = g_i \cdot x_T^{\text{true}}$ under $\sigma(g_i)\vec{a}$.
- **(A4) $\rho(g_i)$ orthogonal:** $\rho(g_i)^\top \rho(g_i) = I$ (norm-preserving on $\mathcal{Z}$).

- **(A5) (closed-loop clause only) equivariant planner + invariant cost:** the planner’s action map satisfies $\pi(\rho(g_i)z, \rho(g_i)z^*) = \sigma(g_i) \pi(z, z^*)$ and the cost is ρ -invariant, $J(\rho(g_i)z; \rho(g_i)z^*) = J(z; z^*)$. (Needed *only* for the closed-loop J identity, not for the prediction-error identity; Step 38 (§4.6) instantiates exactly such a planner.)

Axis 1 — Configuration. Lemma 1 (composition closure). If (A1)–(A4) hold on each generator $g_i \in S$, they hold on every word $w \in \langle S \rangle$, and ρ restricted to $\langle S \rangle$ is an orthogonal-matrix *monoid homomorphism*. *Proof.* Define $\rho(w) := \rho(g_{i_1}) \cdots \rho(g_{i_m})$. For (A2), induct on m : $f(\rho(w)z, \sigma(w)a) = f(\rho(g_{i_1})[\rho(g_{i_2} \cdots)z], \sigma(g_{i_1})[\sigma(g_{i_2} \cdots)a]) = \rho(g_{i_1})f(\rho(g_{i_2} \cdots)z, \sigma(g_{i_2} \cdots)a) = \rho(w)f(z, a)$, using the single-generator (A2) once per step. (A1) and (A3) lift identically by composition; (A4) lifts because a product of orthogonal matrices is orthogonal. $\square$ **Hence k generator checks certify the entire (exponentially large) generated monoid.**

Theorem A (exact configuration certificate). Under (A1)–(A4): for every $w \in \langle S \rangle$, every horizon T , and every action sequence $\vec{a}$,

$$\text{Err}_T(w \cdot x; \sigma(w)\vec{a}) = \text{Err}_T(x; \vec{a}), \quad \text{and for an equivariant planner } J(w \cdot x; w \cdot \text{goal}) = J(x; \text{goal}).$$

Proof. **(i) Rollout equivariance.** By induction on T : $f^{(1)} = f$ is equivariant by (A2)+Lemma 1; assuming $f^{(T)}(\rho(w)z; \sigma(w)\vec{a}) = \rho(w)f^{(T)}(z; \vec{a})$,

$$f^{(T+1)}(\rho(w)z; \sigma(w)\vec{a}) = f(f^{(T)}(\rho(w)z; \sigma(w)\vec{a}), \sigma(w)a_{T+1}) = f(\rho(w)f^{(T)}(z; \vec{a}), \sigma(w)a_{T+1}) = \rho(w)f^{(T+1)}(z; \vec{a}).$$

(ii) Predicted latent at $w \cdot x$. $f^{(T)}(E(w \cdot x); \sigma(w)\vec{a}) \stackrel{(A1)}{=} f^{(T)}(\rho(w)E(x); \sigma(w)\vec{a}) \stackrel{(i)}{=} \rho(w)f^{(T)}(E(x); \vec{a})$. **(iii) Target at $w \cdot x$.** $E((w \cdot x)_T^{\text{true}}) \stackrel{(A3)}{=} E(w \cdot x_T^{\text{true}}) \stackrel{(A1)}{=} \rho(w)E(x_T^{\text{true}})$. **(iv) Cancellation.** Subtracting (iii) from (ii) and using (A4) (an orthogonal ρ preserves the Euclidean norm), $\text{Err}_T(w \cdot x; \sigma(w)\vec{a}) = \|\rho(w)[f^{(T)}(E(x); \vec{a}) - E(x_T^{\text{true}})]\| = \|f^{(T)}(E(x); \vec{a}) - E(x_T^{\text{true}})\| = \text{Err}_T(x; \vec{a})$. The closed-loop identity is the same computation under **(A5)**: the equivariant planner (lifted to words by Lemma 1) selects the $\sigma(w)$ -image of the same actions, and the ρ -invariant cost is unchanged. $\square$

Remark (scope). Theorem A needs **(A3)** — the group must be a symmetry of the *dynamics*, not merely of the encoder. Where the dynamics break the symmetry, the certificate degrades by the amount of that break (Theorem B, §4.5). The error is held *constant across the orbit*, not made small: $\text{Err}_T(x)$ itself can be large (see “flat $\neq$ good”, §6).

Axes 2+3 — Horizon $\times$ Resolution. Theorem B (spectral degradation — a bound, the load-bearing honest result). Relax exactness: suppose the per-generator encoder residual is $\epsilon_{\max} = \max_i \sup_x \|E(g_i \cdot x) - \rho(g_i)E(x)\|$, the per-step predictor error is δ , and the latent map’s Jacobian is, on the ℓ -isotypic channels j , locally diagonalised with multipliers e^{λ_j} (Lyapunov exponents). A Grönwall/Lipschitz accumulation — the word error enters m times (Lemma 1 composes m approximate generators) and the per-step error compounds over T steps amplified by the channel multiplier — gives, for constants c_j depending on the local geometry,

$$|\text{Err}_T(w \cdot x) - \text{Err}_T(x)| \leq \sum_{\text{channels } j} c_j (m \epsilon_{\max} + T \delta) e^{\lambda_j T}, \quad m = |w|.$$

We claim this as a **bound on the form**, not a tight inequality: the c_j are not estimated, and we instead **measure the consequence** — that channel j is certified only to horizon

$$T_j(\epsilon) \sim \frac{1}{\lambda_j} \log \frac{1}{\epsilon} \quad (\lambda_j > 0), \quad T_j = \infty \quad (\lambda_j \leq 0),$$

which §4.2 recovers to within 0.4% (3 seeds). At $\epsilon_{\max}, \delta \rightarrow 0$ (exact equivariance) the configuration term vanishes and Theorem A is recovered.

Corollary (the certified region). $\mathcal{C} = \{(w, T, \epsilon) : |\Delta \text{Err}| \leq \epsilon\}$ is the **coarse** (ϵ large), **slow** ($\lambda_j \leq 0$), **low- $|w|$** corner; its boundary is set by $\langle S \rangle$ (configuration) and the spectrum $\{\lambda_j\}$ (horizon/resolution). With *exact* equivariance ($\epsilon_{\max}, \delta \sim 10^{-6}$ by Theorem A) the configuration axis is unbounded and only the spectrum limits horizon $\times$ resolution — a **trade-off** horizon $\times$ demanded-resolution $\leq \text{const}(\{\lambda_j\})$. This is the formal content of “ultra-long forecasts must be coarse”: eclipses for millennia (conserved $\lambda \leq 0$ actions), weather at $\sim$ two weeks (chaotic $\lambda > 0$).¹

¹An interpretive aside, secondary to the spectral law: a long-range text that asserts only *regime-level* change rather than dated specifics — e.g. the 推背图 / Tuibei-tu tradition — is, structurally, making the *correct* move for a high- λ system, where only the coarse/invariant component is certifiable.

3. The Noether hinge (group $\Rightarrow$ slow)

Theorems A–B hold for *whatever* channels happen to be slow. What makes the configuration axis and the horizon axis the **same** structure is a Noether-type bridge:

Conjecture (group $\Rightarrow$ slow). When the symmetry is a symmetry of the **dynamics** (not merely the encoder), the group-invariant ($\ell=0$) channels coincide with the conserved/slow ($\lambda_j \leq 0$) channels. Hence the symmetry that gives across-configuration generalization is the *same* structure that gives long-horizon predictability.

This is **not automatic** — invariant $\neq$ conserved in general — so we treat it as a *falsifiable, measured* conjecture, with two sharp caveats that keep it honest.

The direction that holds. The defensible statement is the inclusion **slow** $\subseteq$ **invariant** $\oplus$ **conserved-equivariant**, not the converse: the conserved (slowest) modes live in the invariant-or-conserved subspace, but not every invariant channel is slow (a rotation-invariant $|r|^2$ oscillates). The mechanism is Noether’s: a continuous symmetry of the dynamics yields a conserved current. A conserved *scalar* charge ($\ell=0$ Casimir, e.g. energy) **is** group-invariant; a conserved *non-scalar* charge (e.g. 3D angular momentum L , an $\ell=1$ vector) is *equivariant*, **not** invariant. So the slow (conserved) subspace lies in invariant $\oplus$ conserved-equivariant, which **collapses to the invariant isotypic component exactly when every conserved charge is a scalar** — the 2D case below, where the clean slow $\subseteq$ invariant holds literally.

The dimension refinement (measured, §4.3). In 2D (SO(2), Step 50) both conserved quantities — energy E and angular momentum L — are *scalars*, so slow $\subseteq$ invariant literally. In 3D (SO(3), Steps 57–58) angular momentum $L = \sum_i r_i \times v_i$ is a conserved $\ell=1$ **vector**, so the clean 2D containment generalises to

$$\text{slow} \subseteq \text{invariant} \oplus \text{conserved-equivariant},$$

with the equivariant conserved part accessed at *polynomial degree 2* (the cross product). We measure both forms: in 2D the invariant block carries (E, L) at $R^2=0.92\text{--}0.99$ and is exactly group-invariant ($\sim 10^{-16}$, $\forall$ the group; §4.3); in 3D energy is recovered *linearly* from the $\ell=0$ block ($R^2=0.62\text{--}0.91$ across seeds) while L is recovered at $R^2=1.00$ via the degree-2 cross-product readout of the $\ell=1$ block (§4.3). The hinge is thus a *measured* bridge — Noether content dimension-robust, its containment form refined — with the per-seed ranges and gate verdicts reported honestly in §4.3 (the 3D contact gate is INCONCLUSIVE on clean containment; the robust, load-bearing result is the degree-2 L recovery), not assumed.

4. Experiments

All experiments are CPU/1-GPU-scale, seeded, and gated (a run reports INCONCLUSIVE rather than loosen a threshold). Load-bearing results are additionally guarded by unit tests. Code: `experiments/step{47,49,50,51,52}.py`; tests: `tests/test_step{47,50,52}_*.py`.

4.1 Configuration axis — k generators certify an exponential set

Exact-flatness on the embodied model (Step 47, 3 seeds). On a structured object-interaction world with $\text{SE}(3) \times S_n$ symmetry, the equivariant model’s whole-pipeline relMSE is **exactly** $\times 1.00$ **flat over composition words** $m = 0..8$ (constant 0.2625) while a matched non-equivariant MLP blows up by $\times 12.89$. The predictor’s Jacobian spectrum is measured directly: **48 of 96 channels are contractive** ($\sigma \leq 1$, certified long-horizon) vs 48 expansive (max $\sigma = 49$, min $\sigma = 0.003$) — a clean coarse/fine split that is exactly Theorem B’s input. A unit test (`tests/test_step47_certificate.py`) verifies Theorem A holds *architecturally* at initialization (word-deviation 1.2×10^{-7} , no training) while the control deviates 20%.

The exponential demonstration on $\mathbb{Z}_2^6$ (Step 49). The 64 hexagrams of the I Ching are exactly $\{\pm 1\}^6 = \mathbb{Z}_2^6$, with the 6 changing-lines as the generating set. Training a sign-equivariant per-line predictor $f(z, a)_i = h_\theta(a_i) \odot z_i$ on **only the 6 single-line generators** (+ identity, 7 of 64 actions) certifies it over **all** $2^6=64$ **compositions** to machine zero (worst relMSE $\sim 10^{-33}$; equivariance residual exactly 0), while a non-equivariant MLP trained on the same 7 actions degrades monotonically with composition length ($1.6\text{--}1.7 \times 10^{-5}$ seen $\rightarrow 0.59$ at six flips). “ k generator checks $\Rightarrow 2^k$ certified set” made literal, with genuine learning and an honest contrast.

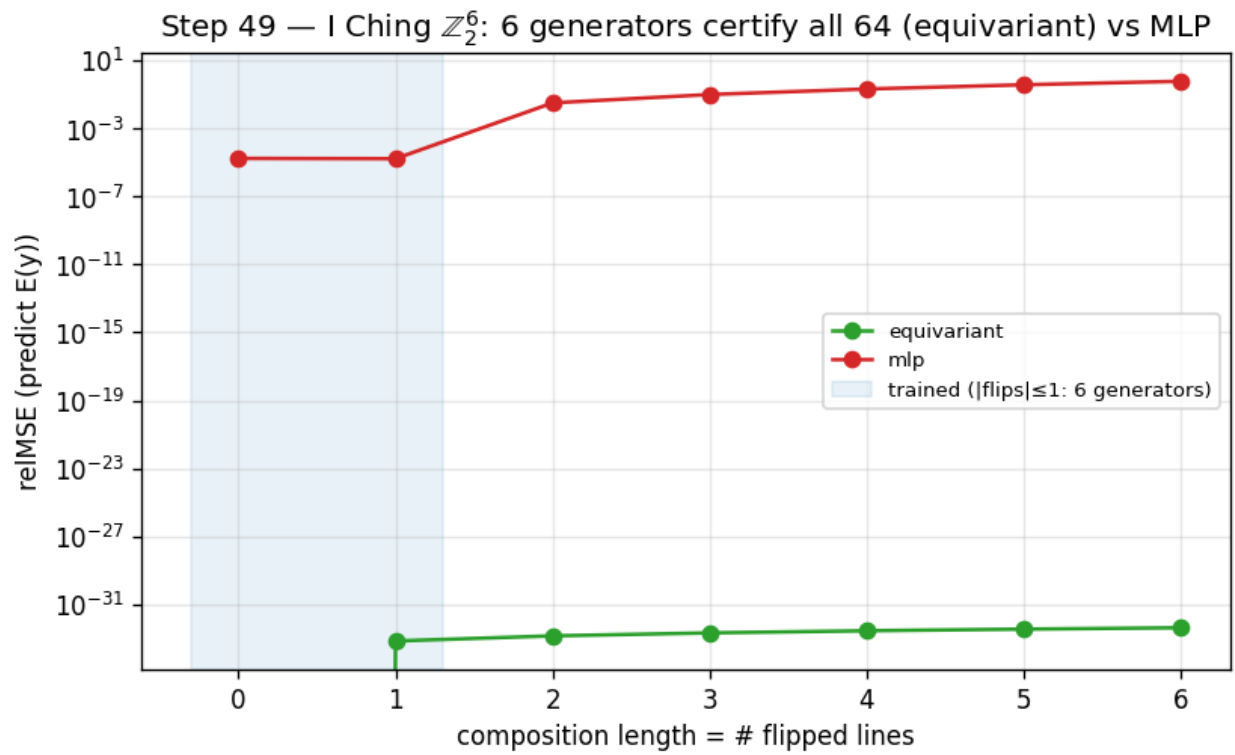


Figure 1: Step 49 — $\mathbb{Z}_2^6$: training on 6 generators certifies all 64 compositions (equivariant, machine zero) while the MLP degrades with composition length.

4.2 Horizon $\times$ resolution — the predictability staircase

Step 52 (3 seeds). A learned one-step predictor on a designed multi-Lyapunov system — a conserved invariant ($\lambda = -\infty$), a contracting channel ($\lambda = \ln 0.75$), a neutral $\text{SO}(2)$ rotor ($\lambda = 0$), and a chaotic angle-doubling channel $z \mapsto z^2$ ($\lambda = \ln 2$) — rolled out autoregressively. The model **recovers the chaotic Lyapunov exponent to within 0.4%** ($\hat{\lambda} = 0.690\text{--}0.692$ across 3 seeds vs $\ln 2 = 0.693$), and its per-channel certified horizon obeys $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$: the chaotic “fine detail” is certifiable only 3–10 steps over the full ϵ grid and its horizon grows *only logarithmically* as ϵ coarsens (measured slope $dT/d\log \epsilon = 1.3\text{--}1.6 \approx 1/\lambda = 1.44$), while the **contracting channel stays certified for the entire 90-step rollout at every ϵ** (the conserved/rotor channels too, except at the finest $\epsilon=0.005$ where they fall to $\sim 40\text{--}90$ steps); at $\epsilon=0.05$ the slow channels outlast the chaotic detail by 12–15 $\times$. This is the predictability staircase — long horizon $\Rightarrow$ coarse + invariant only.

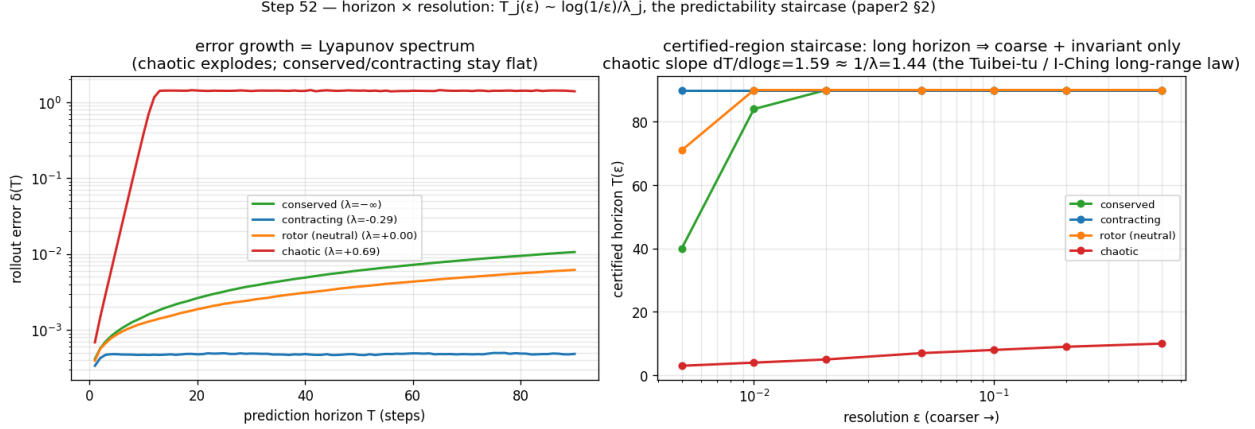


Figure 2: Step 52 — left: rollout error growth = the Lyapunov spectrum; right: the certified-horizon staircase $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$.

4.3 The Noether hinge — slow $\subseteq$ invariant

Step 50 (3 seeds). On a 2D $\text{SO}(2)$ central-force system (conserved energy E and angular momentum L are invariant scalars; the orbital phase is fast), a learned equivariant autoencoder-with-latent-dynamics (a hand-rolled 2D Vector-Neuron model) spontaneously organizes the conserved quantities into its architecturally-invariant ($\ell=0$) block: regression R^2 for (E, L) is 0.92–0.99 **from the invariant block vs ≤ 0.012 from the equivariant ($\ell=1$) block**, and the slowest mode the invariant block admits (0.009–0.031 across seeds) is 4.7–17 $\times$ slower than anything the equivariant block admits (0.145, which equals the orbital phase velocity $2 \sin(\omega \Delta t/2)$). Operationally we read the containment off this min-mode gap — the $\ell=1$ block’s *slowest* achievable mode is already fast, so no slow direction escapes the invariant block in this system — rather than a direct subspace test. The payoff is the certificate: the equivariant model’s slow subspace **is** its invariant subspace, whose out-of-distribution group-action residual is $\sim 10^{-16}$ — *exact and architectural, for all of $\text{SO}(2)$* . A matched MLP also learns E, L ($R^2 = 0.99$) but smears them across directions that **drift by ≈ 1.17** under the same group action, and carries no certificate. The honest non-converse is visible in the data (a fast $|r|^2$ scalar is invariant yet oscillates): **slow $\subseteq$ invariant, not $=$.**

Lift toward embodied/contact, and a principled 3D subtlety (Step 57). Pushing the hinge to a more embodied regime — two bodies in a 3D well with a **soft pairwise repulsion (contact)**, an $\text{SO}(3)$ -equivariant $\ell=0 \oplus \ell=1$ Vector-Neuron model — gives a **partial lift, honestly reported**. The *Noether-content* half lifts: the invariant $\ell=0$ block recovers the conserved $(E, \text{contact-distance})$ at $R^2=0.60\text{--}0.86$ (3 seeds) vs ≤ 0.06 for the $\ell=1$ block, even with 3D contact. But the clean *containment* “slow $\subseteq$ invariant” does **not** lift (invariant slowest mode $0.024 \approx 0.026$ equivariant) — for a principled reason: **in 3D, angular momentum $L = \sum_i r_i \times v_i$ is a conserved $\ell=1$ vector (equivariant)**, so the equivariant block too carries a slow conserved mode. The clean 2D containment (where L is a scalar) becomes, in 3D, **slow $\subseteq$ (invariant $\oplus$ conserved-equivariant)**. The hinge’s Noether-content claim is dimension-robust; its clean-containment form is 2D-specific — a sharpening of scope, not a failure.

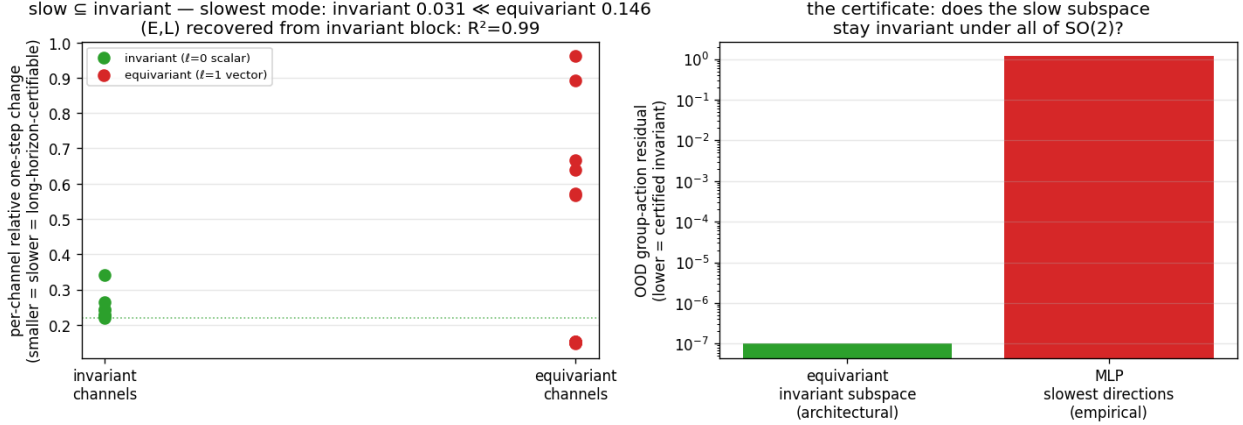


Figure 3: Step 50 — left: the slowest latent modes live in the invariant ($\ell=0$) block; right: the certificate — the invariant subspace stays group-invariant to 10^{-16} where the MLP’s slow directions drift by ~ 1 .

The 3D containment, resolved precisely (Step 58). The conserved physics splits by isotypic *type* and polynomial *degree*: E (an invariant quadratic) is recovered linearly from the $\ell=0$ block ($R^2=0.62$ – 0.91 across seeds), while $L = \sum_i r_i \times v_i$ (a conserved $\ell=1$ vector) is **bilinear** — not linear in either block (≤ 0.04) but recovered at $R^2=1.00$ (range 0.998 – 1.000 , the robust load-bearing result) by the **degree-2 cross-product** readout of the $\ell=1$ block. So $\text{slow} \subseteq (\text{invariant} \oplus \text{conserved-equivariant})$ holds exactly in 3D, with the equivariant conserved part accessed at degree 2 — and that $r \times v$ is exactly the cross-product the companion paper’s degree-1 Vector-Neuron could not form (its tensor-product-message fix), tying the hinge’s 3D form to that paper’s bilinear cap. (The full Step-58 pass-gate, which additionally demands E $R^2 > 0.8$ in the $\ell=0$ block, is met on $1/3$ seeds — surfaced as honestly as Step 57’s INCONCLUSIVE; the degree-2 L recovery is robust across all three.)

4.4 Structure vs scale — scale buys interpolation; structure buys a certificate

Step 51 (3 seeds). Train on a 50° wedge of an $SO(2)$ orbit; test around the full circle. The exactly-equivariant model’s error is **flat over the entire orbit** (out-of-wedge / in-wedge ratio 1.1 – 1.2) — a certificate that holds from partial data by the identity $\text{err}(R_\theta s) = \text{err}(s)$. A non-equivariant MLP **scaled across an $88\times$ parameter range** ($3.8k \rightarrow 337k$) buys real *interpolation* — its in-wedge error drops 31 – $166\times$ and even **beats** the $56\times$ -smaller equivariant model in-distribution — but out-of-wedge the largest MLP remains 10 – $155\times$ worse than the equivariant model (across 3 seeds) and never reaches its floor. Scaling $88\times$ barely moves the out-of-wedge floor; the walls never flatten. The fair nuance — *scale can win in-distribution* — is exactly why the contribution is the **certificate** (an orbit-wide guarantee from partial data), not a per-point accuracy contest.

4.5 Approximate symmetry — graceful degradation and a measured threshold

Step 53 (3 seeds). We break the *world’s* $SO(2)$ symmetry with an anisotropy knob β (the potential becomes $V = \frac{1}{2}(x^2 + (1 + 2\beta)y^2)$, so angular momentum is no longer conserved) and re-run the wedge-train / full-circle-test. Three findings, robust across seeds: (i) at $\beta=0$ the certificate is exact — the equivariant model is 68 – $320\times$ **better out-of-wedge** than the MLP; (ii) as the measured world symmetry-defect ϵ_{world} grows, the equivariant out-of-wedge error grows **smoothly and monotonically** (correlation $+0.88$ – 0.98 — Theorem B’s ϵ_{max} term, a *graceful* slope, not a cliff); and (iii) the equivariant model keeps beating the non-equivariant one up to a **measured symmetry-content threshold** ($\epsilon_{\text{world}} \approx 0.01$ – 0.06), beyond which the (now-wrong) symmetry assumption hurts more than it helps. So *approximate* symmetry buys an *approximate* certificate with exactly the error budget the theory predicts — and the boundary where structure stops paying is itself measured, not assumed.

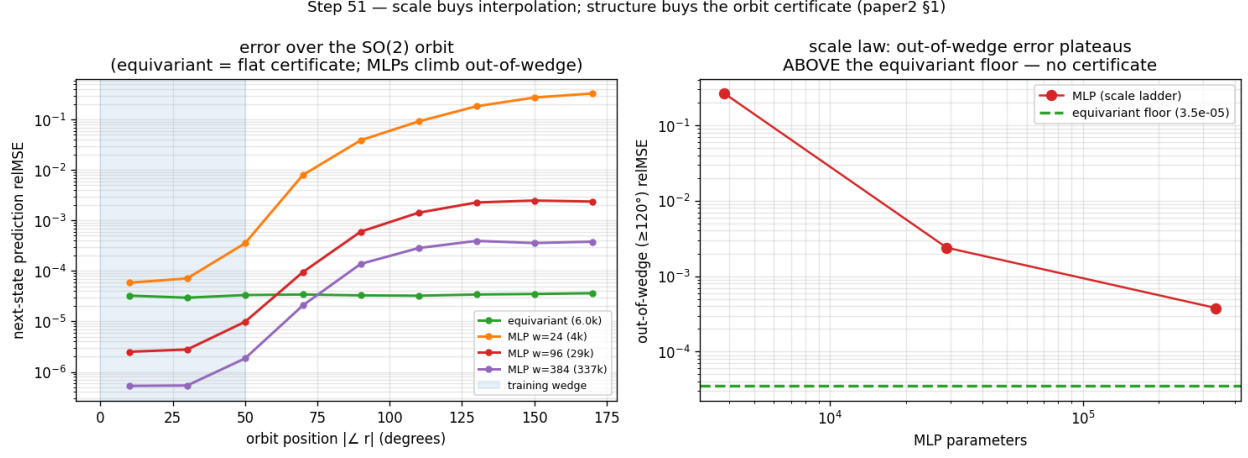


Figure 4: Step 51 — left: error over the SO(2) orbit (equivariant flat; MLPs dip below it in-wedge then climb out); right: out-of-wedge error vs MLP scale plateaus far above the equivariant floor.

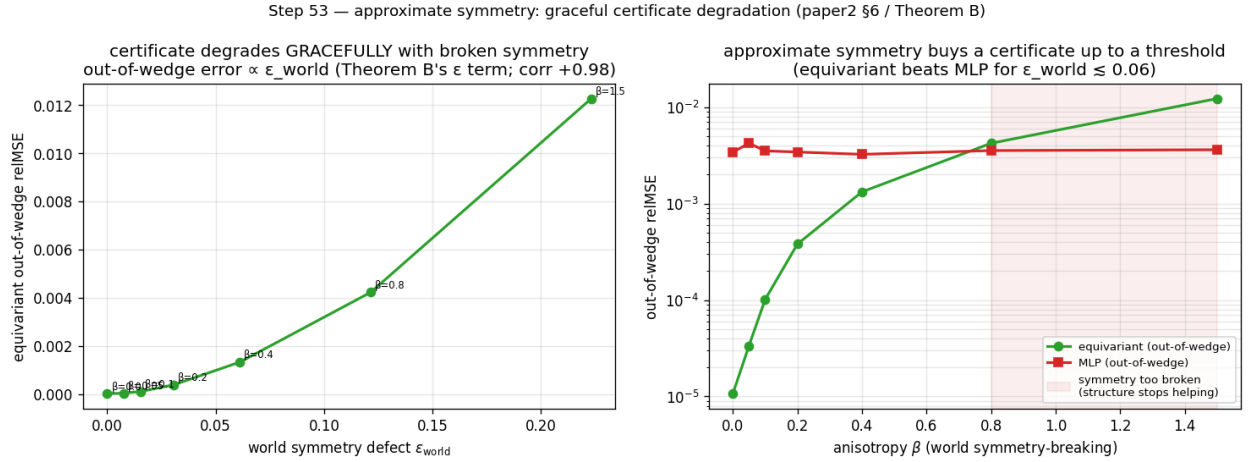


Figure 5: Step 53 — left: equivariant out-of-wedge error grows $\propto \epsilon_{\text{world}}$ (Theorem B's ϵ term); right: the equivariant model beats the MLP out-of-wedge up to a symmetry-content threshold, then crosses over.

4.6 The certificate is actionable — discover the generators, then generate

A certificate is most useful if you need neither know the group in advance nor a decoder to act on it. Two results (re-framed from the companion line) close that loop. **Discovery** (Steps 33, 36): from a blank slate of K learnable 3×3 matrices — nothing antisymmetric or bracket-closing imposed — gradient descent *rediscovers* a frozen teacher’s symmetry algebra ($\mathfrak{so}(3)$, dimension 3, on a rotationally-symmetric teacher; the smaller $\mathfrak{so}(2)_z$, dimension 1, on a rotation-broken one — it reports the *surviving* group and refuses to invent the rest), and **distilling those discovered generators into a free MLP predictor buys back most of the across-group generalization** the hard-wired equivariant model gets for nothing (closing more than half the OOD gap, matching the hand-wired oracle, transferring *exactly* the symmetry discovered). So the generating set S that anchors the configuration axis need not be postulated — it can be *measured*, falsifiably. **Generation** (Step 38): given S , one can *traverse the certified orbit* to an unseen composed goal and execute it **closed-loop without a decoder** — latent-goal reaching driven purely by the equivariance theorem over $K=24$ paired seen-vs-OD SE(3)-orbit tasks, closing ~ 0.59 of the orientation gap on the best deployable variant and — the load-bearing point — achieving the *same* success on the unseen (OOD) orbit as on the seen one (OOD/seen ratio 1.000; the closed-loop $[C]$ guarantee): *exact transfer*, not high absolute success. Together: *discover $S \rightarrow$ certify $\langle S \rangle \rightarrow$ generate (orbit-traverse to an unseen task) $\rightarrow$ act* — the certificate is not merely descriptive but a usable plan-and-execute criterion.

5. Why this is new (precise delta vs prior work)

The $[A]/[B]/[C]$ device (our prior work). A companion line establishes, for *one* group element and a single resolution, that exact equivariance makes one-step error constant across the group $[B]$, with an exact-flatness $[A]$ and a closed-loop invariance $[C]$. **This paper is the lift of that corner along three axes:** Theorem A is the multi-step, full-monoid generalization of $[B]$; Theorem B adds the horizon $\times$ resolution stratification; and the composition-closure lemma turns k generator checks into an exponential certified set. The $[A]/[B]/[C]$ results are the $|w|=1$, single-resolution corner of the certified region.

Equivariant predictors — BRo-JEPA, UWM-JEPA (a corroboration, and a shared caveat). Block-rotation predictors (BRo-JEPA, $\mathbb{Z}/10\mathbb{Z}$ on MNIST, 99.46% zero-shot) and unitary-predictor world models (UWM-JEPA, $U(d)$) independently report striking zero-shot transfer by *matching the predictor to a group structure* — exactly our thesis on different groups. **What they offer us:** cross-group evidence; and our Theorem A explains *why* their zero-shot works (the representation cancels inside the norm — orthogonal for BRo, unitary for UWM) while the closure lemma explains *how far* (the whole generated monoid from the generators — BRo verifies the cyclic case, our Step 49 the exponential $\mathbb{Z}_2^k$ case). **The challenge they raise (which we share and state):** BRo’s circular structure is hand-constructed and its predictor shape matches the answer — the same “constructed-teacher” sweetness as our toy worlds (§6). We do not hide behind their numbers; we cite them as a *mechanism* corroboration, not a scale result.

Latent-geometry regularizers — LeJEPA, and UR-JEPA’s challenge. LeJEPA pushes the latent toward an isotropic Gaussian (SIGReg); **UR-JEPA (Le, 2026) directly challenges that target**, arguing isotropy conflicts with the manifold hypothesis and that a *data-discovered* anisotropy (uniform rectifiability) is better. This is the sharpest question posed to our latent-geometry line, and we answer it head-on (companion paper, and Step 54/56 here): our latent’s anisotropy is neither isotropic (LeJEPA) nor data-discovered (UR-JEPA) but **group-prescribed** — the covariance is *pinned to the representation* ρ ($\rho \Sigma \rho^\top = \Sigma$; measured residual 3×10^{-4} vs a non-equivariant MLP’s 1.04). Crucially, we **concede UR-JEPA its point and sharpen ours:** in a head-to-head (Step 56, single seed, illustrative) the data-fit anisotropy *is* best **in-distribution**, but **only the group-prescribed anisotropy transfers out of distribution** (a $\sim 320\times$ OOD gap, consistent with the 3-seed 68–320 $\times$ of Step 53 and 10–155 $\times$ of Step 51). The certificate is therefore complementary to both — a *first-order, per-situation* guarantee from the group, not a *second-order* distributional prior — and it tells you precisely *when* a discovered anisotropy will fail to generalize (off the training orbit).

Oracle-bypass diagnostics — IMWM (an independent validation, and a complementary bottleneck). IMWM (Gao et al., 2026) uses the *same* falsifiable move as our Steps 43/47 — **replace the learned model with the exact/oracle dynamics and see what is still missing** — and finds a finite-budget planner *still fails*. **What it offers us:** independent corroboration that “swap in the perfect model” is the right instrument. **The distinction (and our response):** IMWM attributes the residual to **search** (proposal-sampling volume); our oracle-bypass attributes it to

representation (permutation-invariant encoder pooling). These are *complementary* bottlenecks, not rivals — a predictability certificate bounds the *model’s* error over $\langle S \rangle \times T \times \epsilon$; it makes no claim about the planner’s search, and IMWM’s result is a useful reminder that a perfect certificate still needs an adequate search budget downstream. We state this scope explicitly (§6).

Geometry on the action side — LDA (an ally on geometry, an opponent on generation). LDA (“The Lie We Tell”, Chuang et al., 2026) names the **Euclidean Fallacy** — flattening an $SE(3)$ pose into $\mathbb{R}^{12}$ breaks the manifold constraint, the coordinate-change equivariance, and geodesic optimality — and fixes it by running diffusion *on* $SE(3)$ (tangent-space score, exp-map retract). **What it offers us:** a strong external statement of our core motivation — *don’t flatten geometric quantities* — now landed on the action/policy side. **The tension it raises:** LDA is a *diffusion policy* (the generative direction this project is contrarian to), so it is simultaneously an ally (geometry) and an opponent (generation); and its gains are “nice, not decisive” (CALVIN task length $3.27 \rightarrow 3.51$, +7.3%), the same *equivariant-method picture* as our Vector-Neuron $\times 1.36$ — modest absolute accuracy, stability under OOD/constraints. Our certificate framing *explains* that picture: a geometric prior buys a *kind* of guarantee (consistency across the group / OOD), not a uniform accuracy jump, so “nice not decisive” average numbers are exactly what one should expect — and the certificate, not the benchmark delta, is the right thing to report.

Predictability horizons (chaos / NWP). The $T(\epsilon) \sim \log(1/\epsilon)/\lambda$ law is classical for dynamical systems; our contribution is to (i) *measure it on a learned latent world model*, (ii) tie its slow subspace to the group-invariant subspace (the Noether hinge), and (iii) fold it into a single multi-axis certificate for equivariant models.

What would falsify the framing. The certificate is not an after-the-fact rationalization: it makes refutable predictions. A *finite, non-equivariant* network that attains exact orbit-flatness ($\text{err}(R \cdot x) = \text{err}(x)$ to machine precision over a full orbit) would break Theorem A’s architectural premise; a symmetric-dynamics system whose conserved/slow modes land *outside* the invariant $\oplus$ conserved-equivariant subspace would break the Noether hinge; and a channel whose certified horizon failed to scale as $\log(1/\epsilon)/\lambda_j$ would break Theorem B. We have not observed these, but they are the experiments that would sink the thesis.

6. Limitations and honesty

- **Toy scale.** All eight experiments are CPU/1-GPU-scale proofs of principle. We claim a *mechanism and a tool*, not a scaled-SOTA benchmark. The certificate’s value is the *kind* of statement it makes, demonstrated cleanly.
- **Theorem B is a spectral bound, not a tightness proof.** Promoting the scalar Lipschitz constant to the Jacobian spectrum $\{e^{\lambda_j}\}$ gives the certified-horizon *form* $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$; it is the load-bearing theory and we **measure its consequences** (Steps 47, 52 recover the spectrum and the law to within 0.4%), but we do not claim the constant c_j or the bound is tight. It is honest as a scaling law, not as an exact inequality. One further scope note: Step 52 measures the *scalar* law on a learned one-step map; the *isotypic* refinement (channels chosen by ℓ -type, which ties the spectrum to equivariance and the Noether hinge) is asserted, not separately measured — Step 47 splits contractive vs. expansive but does not resolve the spectrum by ℓ -type.
- **§4.6 (discovery + generation) re-frames companion-line results** (Steps 33/36/38), not fresh paper2 runs: the discovery/distillation/decoder-free-reach experiments were built for the companion paper and are cited here to show the certificate is *actionable*; they are not new evidence produced for this paper. The decoder-free reach is *exact transfer* (OOD/seen ratio 1.000) at a *partial* absolute success (~ 0.59 of the orientation gap on the best deployable variant); we report the transfer ratio as the load-bearing claim, not absolute task success.
- **Approximate symmetry — measured, with the boundary made explicit.** Real-world symmetry is approximate; §4.5 (Step 53) measures the resulting degradation directly — it is **graceful** (out-of-wedge error $\propto \epsilon_{\text{world}}$, exactly Theorem B’s ϵ_{max} term) and the equivariant certificate remains valuable only **up to a symmetry-content threshold** ($\epsilon_{\text{world}} \approx 0.01\text{--}0.06$), beyond which structure stops paying. The Noether hinge’s **lift to a 3D contact interaction is measured** (§4.3, Steps 57–58): the Noether-content half lifts ($R^2=0.60\text{--}0.86$ across seeds; the clean-containment gate is INCONCLUSIVE), and the 3D containment is now *exact* once stated correctly — slow $\subseteq$ (invariant $\oplus$ conserved-equivariant), with energy linear in the $\ell=0$ block ($R^2=0.62\text{--}0.91$) and angular momentum recovered at $R^2=1.00$ by the degree-2 cross-product of the $\ell=1$ block (Step 58, robust across all 3 seeds). What remains genuinely **not yet done** is lifting this from a clean two-body *toy* contact system to a **real, contact-rich embodied** model (a robot / pixel world), where symmetry is approximate and the latent is learned

end-to-end — the primary next experiment, flagged openly, not hidden.

- **The hinge is a measured conjecture.** Confirmed on a controlled system, with the honest non-converse (invariant $\nRightarrow$ slow). We do not claim a proof that learning *must* place conserved quantities in the invariant block — only that, where the dynamics are symmetric, it *does* in our experiments.
- **“Flat” is not “good.”** A constant error across the group says nothing about the error’s *magnitude*; Theorem A certifies *consistency*, and we report absolute errors alongside ratios throughout.

Sharpest objections, answered

We stress-tested the claims adversarially; the strongest objections and our honest answers:

1. **“Assumption (A3) is the whole game and rarely holds exactly.”** *Correct, and it is the paper’s stated scope, not a hidden step.* The **exact** certificate (Theorem A) holds precisely where the group is a symmetry of the *dynamics* — orbital/conservative systems, free space, idealised manipulation. Everywhere else one is in Theorem B’s regime, and our contribution there is to **quantify** the degradation: §4.5 (Step 53) shows it is *graceful* (out-of-wedge error $\propto \epsilon_{\text{world}}$) up to a measured symmetry-content threshold ($\epsilon_{\text{world}} \approx 0.01\text{--}0.06$, seed-dependent — one seed crosses over as early as $\epsilon \approx 0.008$), beyond which structure stops paying. The honest reading is “exact on symmetric dynamics; gracefully approximate, with a measured boundary, elsewhere” — which is *more* useful than an unconditional claim.
2. **“Theorem A is just equivariance restated.”** The *one-element* fact (equivariant model $\Rightarrow$ orbit-constant error) is prior work’s corner. The contribution is the **certified set**: the closure lemma turns k generator checks into the whole exponential monoid $\langle S \rangle$ (Step 49: $6 \rightarrow 64$), the **multi-step** lift to any horizon, the **horizon $\times$ resolution** stratification (Theorem B), and the **Noether hinge** unifying the configuration and time axes. A guarantee over an exponential region across three axes is not the single-element statement.
3. **“Scale was only swept to 337k parameters.”** True — the out-of-wedge floor *plateaus* over the $88\times$ we test (Step 51). The “scale never reaches the certificate” claim rests on an **architectural** fact, not an infinite sweep: no finite non-equivariant network satisfies the exact identity $\text{err}(R \cdot x) = \text{err}(x)$, so its orbit-flatness can shrink with scale but never reach the equivariant floor by construction. We demonstrate the plateau empirically and argue the limit structurally.
4. **“The hinge is a conjecture ‘confirmed’ on systems you built.”** We label it a *measured conjecture*, never a theorem. The test is non-trivial — the model *learns* (not hand-sets) where to place the conserved quantities, and the honest non-converse ($|r|^2$ invariant-but-fast) plus the 3D refinement (Step 58) rule out a tautology — but it is shown only on constructed symmetric systems; lifting it to a *real* embodied model is open.
5. **“The toy testbeds (incl. the I Ching framing) are rigged / gimmicky.”** The equivariant architecture is *designed* to match the problem’s structure — that is the thesis, and the MLP control plus genuine learning make each a clean existence proof, not a fit-to-the-answer. The 推背图 / celestial-mechanics reading is an interpretive aside (clearly secondary to the math), kept because it names the horizon-resolution law memorably.

7. Conclusion

For equivariant world models, structure delivers a *kind* of result scaling cannot: a **predictability certificate** — a provable, computable, training-free region across configuration, horizon, and resolution. We established the master theorem (Theorem A — the exact configuration certificate — proved, its closed-loop clause under the assumed equivariant-planner condition (A5); Theorem B — the spectral horizon $\times$ resolution law — stated as a bound and measured), exhibited the certified region as the coarse-invariant-slow-low-composition corner, measured the Noether hinge that unifies the symmetry and time axes, and showed that an $88\times$ -scaled non-equivariant model buys interpolation but never the certificate — an *architectural* impossibility (no finite non-equivariant net satisfies $\text{err}(R \cdot x) = \text{err}(x)$), not merely an unfinished sweep. *Scale buys interpolation; structure buys a certificate.* The same criterion that tells you which compositions a robot policy will handle zero-shot also tells you why eclipses are forecastable for millennia and weather is not — one structural law, read across domains.

References

Concurrent and prior work referenced above (arXiv identifiers from the project’s source notes; external numbers are quoted from the cited works):

- **BRO-JEPA** — block-rotation $\mathbb{Z}/10\mathbb{Z}$ predictor; 99.46% best-config zero-shot on MNIST modular arithmetic. arXiv:2606.01372.
- **UWM-JEPA** — unitary-predictor ($U(d)$) belief-space world model. arXiv:2605.25313.
- **UR-JEPA** (Le, 2026) — uniform-rectifiability latent regularizer; manifold-hypothesis critique of isotropic SIGReg. arXiv:2606.01443.
- **IMWM** (Gao et al., 2026) — oracle-bypass diagnosis localizing the residual to planner *search*. arXiv:2606.01626.
- **LDA / “The Lie We Tell”** (Chuang et al., 2026) — the *Euclidean Fallacy*; SE(3)-intrinsic diffusion policy; CALVIN average task length $3.27 \rightarrow 3.51$ (+7.3%). arXiv:2606.01847.
- **LeJEPA** (Balestriero & LeCun, 2025) — SIGReg / isotropic-Gaussian self-supervised objective (the latent-geometry target that UR-JEPA, and our group-prescribed anisotropy, depart from).

The $T(\epsilon) \sim \log(1/\epsilon)/\lambda$ predictability-horizon law is classical (Lyapunov / Lorenz; standard numerical-weather-prediction practice).

Reproducibility

Result	Experiment	Test	Seeds	Headline number
Exact-flatness + spectrum	experiments/step47_certificate.py	tests/test_step47_certificate.py	3	relMSE $\times 1.00$ flat ($m=0.8$); 48/96 contractive
$\mathbb{Z}_2^6$ exponential cert	experiments/step49_— iching_certificate.py	—	1	6 generators $\rightarrow$ all 64; worst relMSE $\sim 10^{-33}$
Noether hinge	experiments/step50_noether_hinge.py	tests/test_step50_noether_hinge.py	3	R^2 0.92–0.99 vs ≤ 0.012 ; cert 10^{-16} vs 1.17
Structure vs scale	experiments/step51_— structure_vs_scale.py	—	3	equiv flat 1.1–1.2; in-wedge gain 31–166 $\times$; best MLP 10–155 $\times$ above equiv floor
Horizon $\times$ resolution	experiments/step52_horizon_resolution.py	tests/test_step52_horizon_resolution.py	3	out-of-wedge $\hat{\lambda}=0.69$ vs $\ln 2$; slope $\approx 1/\lambda$
Approximate symmetry	experiments/step53_— approximate_symmetry.py	—	3	cert exact at $\beta=0$ (68–320 $\times$); graceful $\propto \epsilon$ (corr 0.88–0.98); threshold $\epsilon \approx 0.01$ –0.06

Result	Experiment	Test	Seeds	Headline number
Embodied/contact hinge lift	experiments/step57_ embodied_ hinge.py		3	gate INCONCLUSIVE on clean containment (slow_{inv} 0.024–0.038 $\approx$ slow_{eq} 0.026–0.059); Noether content half lifts (invariant $R^2=0.60$ –0.86 vs ≤ 0.06); resolved by Step 58
3D-aware containment	experiments/step58_ 3d_ containment.py		3	$E \rightarrow \ell=0$ linear ($R^2=0.62$ –0.91); L bilinear $\rightarrow \ell=1$ degree-2 cross ($R^2=1.00$, range 0.998–1.000); slow $\subseteq$ (invariant $\oplus$ conserved- equivariant)

Every experiment sets random seeds explicitly, prints an INCONCLUSIVE verdict rather than loosen a gate, and writes its figure + JSON to papers/figures/. **Multi-seed reproducibility:** Steps 50/51/52/53/57/58 commit per-seed JSONs papers/figures/step5*_seeds.json (seeds 0/1/2), regenerated by experiments/aggregate_seeds.py, and every range quoted from them is the seed min–max; Step 47 self-aggregates its 3 seeds into the means reported in step47_certificate.json (no per-seed range committed; §4.1 quotes no range). The full test suite (81 tests) passes together; tests/conftest.py isolates the float64 experiments from the float32 codebase.